An Introduction into Random Walk and various problems

Random walk is a process in which an object moves through a sequence of positions, with each step determined by a random outcome. Due to the random nature of movements in random walk various interesting phenomena may appear. For instance, imagine a random walker that moves either 1 unit up, or down or to the left, or right every second. What is the probability that he will eventually return to the origin? A secondary question would be for a simple random walk in which the walker can only move one unit left or right every second, what is the average distance between the origin and the walker.

These two problems are rather well-known random walk problems. For the first problem, we know that the random walker can only appear on an even number of steps. Denote the probability of the random walker being at the origin at time $2t$ by P_{2t}.

We see that there are a total of $4^{2t} = 16^t$ different combinations of total paths to take. In order for the walker to return back to the origin, we know that the number of left and right steps must be equal, as well as the number of up and down steps.

Given that (k) is the number of left steps, we know there must be (k) right steps and (t-k) up steps and (t-k) down steps. The number of ways for this to occur is thus $\frac{(2t)!}{k!^2(t-k)!^2}$.

We see that the denominator is related to the denominator of

$$\binom{t}{k}^2.$$

This motivates us to try to find a potential simplification to this expression. We then note that

$$\frac{(2t)!}{t!t!} \cdot \frac{t!}{k!(t-k)!}^2$$

is the same as our previous expression. This gives us

$$\binom{2t}{t}\binom{t}{k}^2.$$

We note that by Vandermonde's identity, this gives us

$$\binom{2t}{t}^2.$$ As such, the probability that the random walker eventually reaches the origin at time (2t) is equal to

$$\frac{\binom{2t}{t}^2}{16^t}.$$

We can use Stirling's approximation to approximate $\binom{2t}{t} \approx \frac{4^t}{\sqrt{\pi t}}.$ This gives us the final summative expression

$$\sum_{t=1}^{\infty} P_{2t} \approx \sum_{t=1}^{\infty} \frac{1}{\pi t},$$

which we know is equal to infinity due to the divergence of the harmonic series. Since the sum is infinite, we know that the random walker will eventually hit the origin with probability 1.

We now move onto the second problem. Given a one dimensional graph of a random walker moving. What is the average distance between the walker and the origin.

We seek to find the bound mathematically.

Let S_n denote the position of a one-dimensional random walker after n steps, starting at the origin. At each step, the walker moves one unit to the right or left, each with probability $\frac{1}{2}$.We want to determine the expected distance between the walker and the origin:

$$a_n = [|S_n|].$$

Since each step is independent, we can write

$S_n = X_1 + X_2 + \cdots + X_n$, where $X_i = \pm 1$

Let S_n represent the position of the random walker after n steps, and let a_n represent the expected distance between the walker and the origin after n steps. Thus,

$$a_n = E[|S_n|].$$

We want to find a recurrence relation expressing a_{n+1} in terms of a_n.

Since the walker moves either one unit to the right or one unit to the left with equal probability, if the walker is currently at position x, its position after the next step will be either $x + 1$ or $x - 1$. Therefore, its expected distance from the origin after the next step is

$$\mathbb{E}[|S_{n+1}| \mid S_n = x] = \frac{|x+1| + |x-1|}{2}.$$

We consider two cases depending on whether the walker is at the origin.

Case 1: The walker is not at the origin

Suppose $x \neq 0$. We consider the cases $x > 0$ and $x < 0$.

If $x > 0$, then

$$\begin{aligned}
\mathbb{E}[|S_{n+1}| \mid S_n = x] &= \frac{|x+1| + |x-1|}{2} \\
&= \frac{(x+1) + (x-1)}{2} \\
&= \frac{2x}{2} \\
&= x \\
&= |x|.
\end{aligned}$$

Similarly, if $x < 0$, then

$$\begin{aligned}\mathbb{E}[|S_{n+1}| \mid S_n = x] &= \frac{|x+1| + |x-1|}{2} \\ &= \frac{(-x-1) + (-x+1)}{2} \\ &= \frac{-2x}{2} \\ &= -x \\ &= |x|.\end{aligned}$$

Therefore, whenever the walker is not at the origin, its expected distance after the next step equals its current distance:

$$\mathbb{E}[|S_{n+1}| \mid S_n = x] = |x|, \qquad x \neq 0.$$

This happens because the walker has an equal probability of moving one unit closer to the origin or one unit farther away from it.

Case 2: The walker is at the origin

If $x = 0$, the walker moves to either 1 or -1. In both cases, its distance from the origin becomes 1. Hence,

$$\begin{aligned}\mathbb{E}[|S_{n+1}| \mid S_n = 0] &= \frac{|0+1| + |0-1|}{2} \\ &= \frac{1+1}{2} \\ &= 1.\end{aligned}$$

Thus, when the walker is at the origin, its expected distance increases from 0 to 1.

The preceding results show that the expected distance after the next step equals the current distance, with an additional 1 whenever the walker is currently at the origin.

Define the indicator function

$$\mathbf{1}_{\{S_n=0\}} = \begin{cases} 1 & \text{if } S_n = 0, \\ 0 & \text{if } S_n \neq 0. \end{cases}$$

We can then express the result as

$$\mathbb{E}[|S_{n+1}| \mid S_n] = |S_n| + \mathbf{1}_{\{S_n=0\}}.$$

Taking expectations on both sides gives

$$\begin{aligned}\mathbb{E}[|S_{n+1}|] &= \mathbb{E}\left[|S_n| + \mathbf{1}_{\{S_n=0\}}\right] \\ &= \mathbb{E}[|S_n|] + \mathbb{E}[\mathbf{1}_{\{S_n=0\}}].\end{aligned}$$

By definition,

$$E[|S_{n+1}|] = a_{n+1}$$

and

$$E[|S_n|] = a_n.$$

Furthermore, the expected value of an indicator function equals the probability of the event it represents. Therefore,

$$E[1_{\{S_n=0\}}] = P(S_n = 0).$$

Substituting these expressions into the previous equation yields

$$a_{n+1} = a_n + P(S_n = 0).$$

Hence, the recurrence relation is

$$\boxed{a_{n+1} = a_n + \mathbb{P}(S_n = 0)}.$$

Finding the general expression from the recurrence

Since the walker starts at the origin, its initial distance is zero:

$$a_0 = 0.$$

Applying the recurrence repeatedly gives

$$
\begin{aligned}
a_1 &= a_0 + \mathbb{P}(S_0 = 0),\\
a_2 &= a_1 + \mathbb{P}(S_1 = 0),\\
a_3 &= a_2 + \mathbb{P}(S_2 = 0),\\
&\vdots\\
a_n &= a_{n-1} + \mathbb{P}(S_{n-1} = 0).
\end{aligned}
$$

Substituting each preceding equation into the next, we obtain

$$
\begin{aligned}
a_n = a_0 &+ \mathbb{P}(S_0 = 0) + \mathbb{P}(S_1 = 0) + \cdots\\
&+ \mathbb{P}(S_{n-1} = 0).
\end{aligned}
$$

Since $a_0 = 0$, this simplifies to

$$
\boxed{a_n = \sum_{j=0}^{n-1} \mathbb{P}(S_j = 0)}.
$$

Therefore, the expected distance from the origin after n steps equals the sum of the probabilities that the walker is at the origin at each time from 0 through $n - 1$. This completes the proof of the recurrence relation and provides the expression needed to calculate the expected distance in the following steps.

We established that

$$
a_n = \sum_{j=0}^{n-1} P(S_j = 0).
$$

We now need to calculate the probability that the walker is at the origin after j steps.

The walker can return to the origin only after an even number of steps because each step changes the parity of the walker's position. Therefore, if the walker takes an odd number of steps, the probability of being at the origin is zero:

$$
P(S_{2k+1} = 0) = 0.
$$

Suppose the walker takes $2k$ steps. To return to the origin, the walker must take exactly k steps to the right and k steps to the left.

The number of ways to arrange these steps is

$$\binom{2k}{k} = \frac{(2k)!}{k!k!}.$$

Since each step independently has probability $\frac{1}{2}$ of moving in either direction, every sequence of $2k$ steps has probability

$$\left(\frac{1}{2}\right)^{2k} = \frac{1}{4^k}.$$

Consequently, the probability of returning to the origin after $2k$ steps is

$$\boxed{\mathbb{P}(S_{2k} = 0) = \frac{\binom{2k}{k}}{4^k}}.$$

Substituting these probabilities into the expression obtained in Step 2, we get

$$a_n = \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} \frac{\binom{2k}{k}}{4^k}.$$

This gives an exact expression for the expected distance from the origin after n steps.

We will now prove that the summation can be simplified to

$$\boxed{a_n = \frac{n}{2^{n-1}} \binom{n-1}{\lfloor (n-1)/2 \rfloor}} \qquad n \geq 1.$$

To prove this formula, define

$$b_n = \frac{n}{2^{n-1}} \binom{n-1}{\lfloor (n-1)/2 \rfloor}.$$

We will show that b_n satisfies the same recurrence relation as a_n:

$$a_{n+1} - a_n = P(S_n = 0).$$

First, for $n = 1$,

$$b_1 = \frac{1}{2^0}\binom{0}{0} = 1.$$

The walker must be one unit away from the origin after one step, so $a_1 = 1 = b_1$.

Now consider even values of n. Let $n = 2k$. Then

$$b_{2k} = \frac{2k}{2^{2k-1}}\binom{2k-1}{k-1}$$

and

$$b_{2k+1} = \frac{2k+1}{2^{2k}}\binom{2k}{k}.$$

Using the binomial coefficient identity

$$\binom{2k-1}{k-1} = \frac{1}{2}\binom{2k}{k},$$

we obtain

$$\begin{aligned} b_{2k+1} - b_{2k} &= \frac{2k+1}{2^{2k}}\binom{2k}{k} - \frac{2k}{2^{2k}}\binom{2k}{k} \\ &= \frac{\binom{2k}{k}}{2^{2k}} \\ &= \frac{\binom{2k}{k}}{4^k}. \end{aligned}$$

From Step 3, we know that

$$P(S_{2k} = 0) = \frac{\binom{2k}{k}}{4^k}.$$

Therefore,

$$b_{2k+1} - b_{2k} = P(S_{2k} = 0).$$

Next, consider odd values of n. Let $n = 2k + 1$. We have

$$b_{2k+1} = \frac{2k+1}{2^{2k}}\binom{2k}{k}$$

and

$$b_{2k+2} = \frac{2k+2}{2^{2k+1}}\binom{2k+1}{k}.$$

Using the identity

$$\binom{2k+1}{k} = \frac{2k+1}{k+1}\binom{2k}{k},$$

we obtain

$$\begin{aligned} b_{2k+2} &= \frac{2k+2}{2^{2k+1}} \cdot \frac{2k+1}{k+1}\binom{2k}{k} \\ &= \frac{2k+1}{2^{2k}}\binom{2k}{k} \\ &= b_{2k+1}. \end{aligned}$$

Thus,

$$b_{2k+2} - b_{2k+1} = 0.$$

Because the walker cannot return to the origin after an odd number of steps,

$$P(S_{2k+1} = 0) = 0.$$

Consequently,

$$b_{2k+2} - b_{2k+1} = P(S_{2k+1} = 0).$$

We have shown that b_n satisfies the same recurrence relation as a_n for both even and odd values of n. Since they also have the same initial value, mathematical induction proves that $a_n = b_n$ for all $n \geq 1$.

Therefore, the exact expected distance is $\boxed{a_n = \frac{n}{2^{n-1}} \binom{n-1}{\lfloor (n-1)/2 \rfloor}}$.

Although the exact formula gives the expected distance for any number of steps, we want to understand how the distance grows as n becomes large.

We use Stirling's approximation:

$$m! \sim \sqrt{2\pi m}\left(\frac{m}{e}\right)^m.$$

Applying this approximation to the central binomial coefficient yields

$$\binom{2k}{k} \sim \frac{4^k}{\sqrt{\pi k}} ask \to \infty.$$

Consider an even number of steps, $n = 2k$. From Step 4, we have

$$a_{2k} = \frac{2k}{2^{2k-1}}\binom{2k-1}{k-1}.$$

Using the identity

$$\binom{2k-1}{k-1} = \frac{1}{2}\binom{2k}{k},$$

we can rewrite the expression as

$$a_{2k} = \frac{2k}{4^k}\binom{2k}{k}.$$

Substituting the central binomial coefficient approximation gives

$$a_{2k} \sim \frac{2k}{4^k} \cdot \frac{4^k}{\sqrt{\pi k}}.$$

Canceling the powers of 4, we obtain

$$a_{2k} \sim \frac{2k}{\sqrt{\pi k}} = 2\sqrt{\frac{k}{\pi}}.$$

Since $n = 2k$, we have $k = \frac{n}{2}$. Substituting gives

$$a_n \sim 2\sqrt{\frac{n}{2\pi}}$$
$$= \sqrt{\frac{2n}{\pi}}.$$

This result also holds for odd values of n. Indeed, the recurrence established in Step 2 shows that

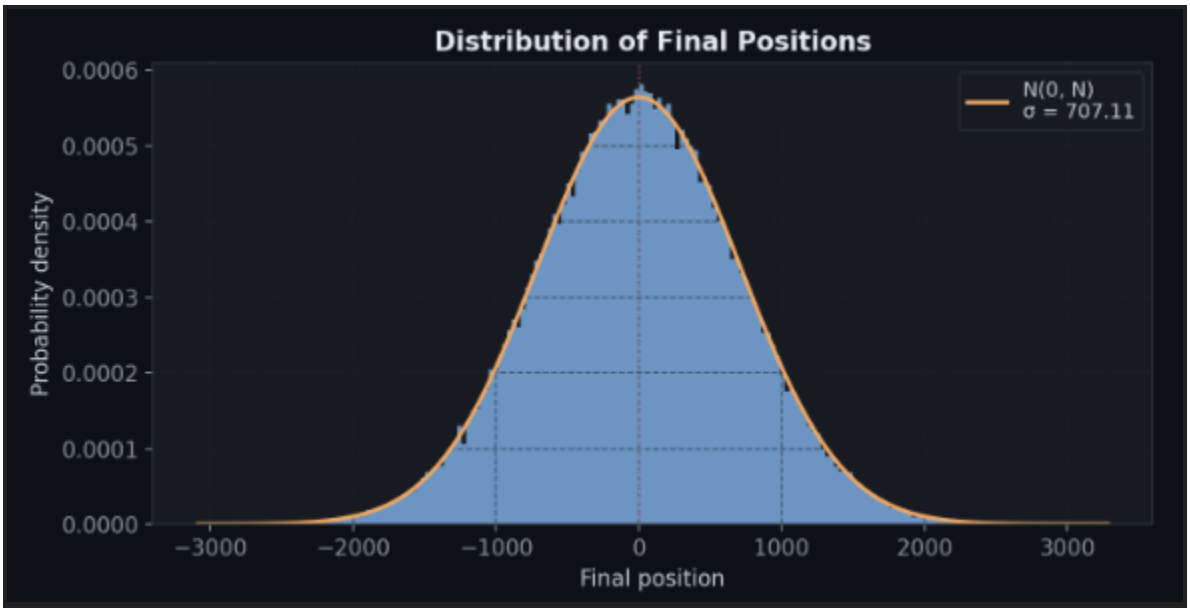
$$a_{2k+2} = a_{2k+1},$$

because the probability of being at the origin after an odd number of steps is zero. Since consecutive step counts become relatively indistinguishable as n grows, the same asymptotic approximation applies to odd n.

Therefore,

$$\boxed{a_n \sim \sqrt{\frac{2n}{\pi}}}.$$

We can run a monte carlo simulation to give us an estimate of roughly the average distance found testing for 500,000 steps and 200,000 simulations.Which gives us the following graph.

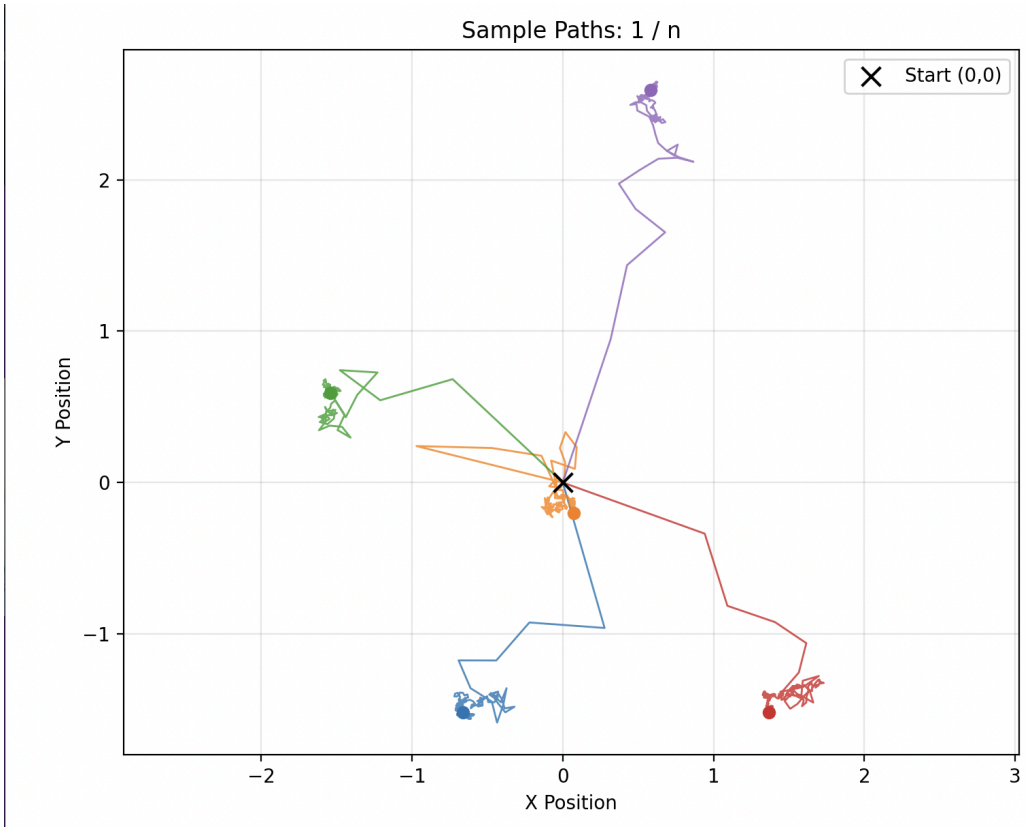


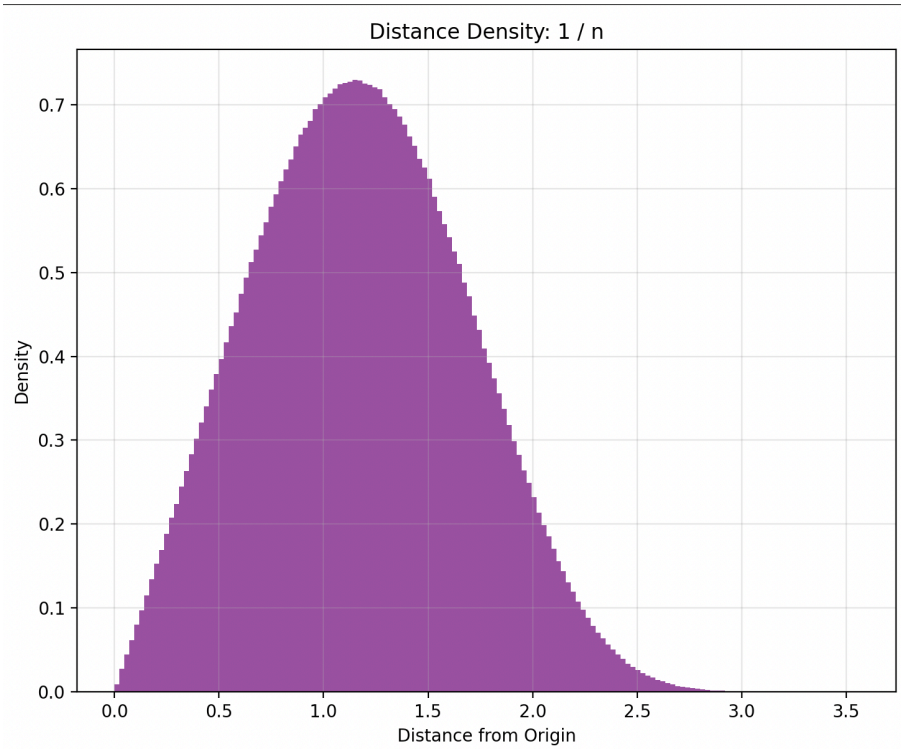
This estimation fits in nicely with our expected estimation of $\sqrt{\frac{1000000}{\pi}} \approx 564$ whereas our real average is closer to 562 in the simulation likely due to variance.

Now imagine a random walker who is not only staggering but also tiring. In contrast to a simple random walker who takes steps of equal length but random direction, this random walker first takes a step of length 1, then a step of length 1/2, then a step of length 1/3, and so on. As in a simple random walk the direction of each step is equally likely to be left as right and independent of the other steps. We know that the sum of all these steps is still equal to infinity since the

harmonic series diverges. However the question now remains? Is it guaranteed that the random walker will return home. Additionally, will the random walker approach a given point?

To begin we can first try to simulate 10,000,000 runs each with adding up the harmonics to 1/20000. The results are as shown.





We see from these two graphs that the path of the harmonics seems to eventually converge to a random point. Furthermore as noted by the second graph the distance between the origin and the eventual end point is close to around 1.25 and the graph seems to form a bell curve like shape. Using this new information we now try to find a proof as to why the random walks of 1/n converge at a single point. We can use Kolmogorov's convergence theorem to justify why this is the case.

First we give a proof of Kolmogorov's convergence theorem

Claim. If $X_1, X_2, \cdots$ are independent, mean zero, and $\sum_k \mathrm{Var}(X_k) < \infty$, then $S_n = \sum_{k \le n} X_k$ converges almost surely.

The proof has two steps. First, a bound on how far the partial sums can wander. Second, a check that they eventually stop wandering.

Step 1: Kolmogorov's inequality

For any $\lambda > 0$,

$$\mathbb{P}\Big(\max_{k\le n}|S_k|\ge\lambda\Big)\le\frac{\mathrm{Var}(S_n)}{\lambda^2}.$$

Let A_k be the event that k is the first index with $|S_k| \geq \lambda$. These events are disjoint, and their union is the event in question. Write $S_n = S_k + (S_n - S_k)$. On A_k, the term $S_k \mathbf{1}_{A_k}$ depends only on $X_1, \cdots, X_k$, while $S_n - S_k$ depends only on later variables and has mean zero. The cross term therefore vanishes:

$$\mathbb{E}[S_n^2 \mathbf{1}_{A_k}] = \mathbb{E}[S_k^2 \mathbf{1}_{A_k}] + \mathbb{E}[(S_n - S_k)^2 \mathbf{1}_{A_k}] \;\geq\; \mathbb{E}[S_k^2 \mathbf{1}_{A_k}] \;\geq\; \lambda^2\, \mathbb{P}(A_k).$$

Summing over k gives $$\mathrm{Var}(S_n) = \mathbb{E}[S_n^2] \geq \sum_k \mathbb{E}[S_n^2 \mathbf{1}_{A_k}] \geq \lambda^2\, \mathbb{P}(\max_k |S_k| \geq \lambda)$$.

Fix m and apply Step 1 to the shifted sums $S_k - S_m$ for $k > m$. Letting $n \to \infty$,

$$\mathbb{P}\Big(\sup_{k>m}|S_k - S_m|\ge\varepsilon\Big)\le\frac{1}{\varepsilon^2}\sum_{j>m}\mathrm{Var}(X_j).$$

The right side is the tail of a convergent series, so it tends to 0 as $m \to \infty$.

Now let $M_m = \sup_{k,l>m}|S_k - S_l|$. By the triangle inequality, $M_m \leq 2\sup_{k>m}|S_k - S_m|$, so $\mathbb{P}(M_m \geq 2\varepsilon) \to 0$. Since M_m decreases as m grows, it has a limit $M_\infty \geq 0$, and

$$\mathbb{P}(M_\infty \geq 2\varepsilon) \leq \mathbb{P}(M_m \geq 2\varepsilon) \to 0 \quad \text{for every } \varepsilon > 0.$$

Hence $M_\infty = 0$ almost surely. That says the gaps between far-out partial sums shrink to zero, so (S_n) is a Cauchy sequence, and therefore converges.

Proof that the walker converges to a point

We will assume that the walker starts at the origin, takes steps of length $1/n$ at step n, and chooses each direction independently and uniformly at random. We want to prove that the walk converges to a single point in the plane with probability 1.

Let S_n denote the walker's position after n steps. Its position is

$$S_n = \sum_{k=1}^{n} X_k,$$

where the k-th step is

$$X_k = \frac{1}{k} \begin{pmatrix} \cos \theta_k \\ \sin \theta_k \end{pmatrix},$$

and the angles θ_k are independent and uniformly distributed between 0 and 2π.

Each step has length $1/k$, so the walker's total distance traveled is

$$\sum_{k=1}^{\infty} \frac{1}{k}.$$

This sum diverges, meaning the total distance traveled is infinite. However, because the directions are random, the steps can cancel one another, allowing the walker's position to converge.

Let C be the event that the walk converges:

$$C = \left\{ \lim_{n \to \infty} S_n exists \right\}.$$

Changing any finite number of steps only shifts the walker's later positions by a fixed vector. Such a change cannot affect whether the walk converges.

Therefore, the event C depends only on the directions of steps arbitrarily far into the future. This makes C a tail event.

By Kolmogorov's 0–1 law,

$$\mathbb{P}(C) \in \{0, 1\}$$

Thus, the walk either converges with probability 1 or fails to converge with probability 1. We must now establish which possibility occurs.

Consider the horizontal and vertical coordinates separately:

$$S_n = (S_n^{(x)}, S_n^{(y)}).$$

The horizontal coordinate is

$$S_n^{(x)} = \sum_{k=1}^{n} \frac{\cos\theta_k}{k},$$

and the vertical coordinate is

$$S_n^{(y)} = \sum_{k=1}^{n} \frac{\sin\theta_k}{k}.$$

Because the directions are uniform and independent,

$E[\cos\theta_k] = 0, \qquad \mathbb{E}[\sin\theta_k] = 0$ Also, $\operatorname{Var}(\cos\theta_k) = \operatorname{Var}(\sin\theta_k) = \frac{1}{2}$

Consequently, the variance of each coordinate's k-th increment is

$$\operatorname{Var}\left(\frac{\cos\theta_k}{k}\right) = \frac{1}{2k^2}$$

and similarly for the vertical coordinate.

The sum of these variances is finite:

$$\sum_{k=1}^{\infty} \frac{1}{2k^2} = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12} < \infty.$$

By Kolmogorov's convergence theorem for independent random variables, a series of independent, mean-zero random variables whose variances have a finite sum converges almost surely.

Therefore, both coordinate series converge almost surely:

$S_n^{(x)} \longrightarrow S_\infty^{(x)}$ and $S_n^{(y)} \longrightarrow S_\infty^{(y)}$.

Since both coordinates converge, the two-dimensional position also converges:

$S_n \longrightarrow S_\infty = \left(S_\infty^{(x)}, S_\infty^{(y)}\right)$ with probability 1.

We established that the probability of convergence must be either 0 or 1and that the walk actually converges almost surely. Therefore, the probability cannot be 0, and we conclude that

$$\mathbb{P}\left(\lim_{n\to\infty} S_n \text{ exists}\right) = 1$$

Hence, the two-dimensional random walk with independently chosen uniform directions and step lengths $1/n$ converges almost surely to a single, finite, random point in the plane.

Random Walks as a Method for Generating Random Numbers

A random walk with decreasing step lengths can converge almost surely to a random limit point. This means we can use a sequence of fair coin flips to generate a random number. The sizes of the steps determine how spread out the resulting numbers are, how they are bounded, and how closely their distribution resembles a normal distribution.

Let $\varepsilon_1, \varepsilon_2, \cdots$ be independent random variables that each equal $+1$ or -1 with probability $1/2$. Define the step sizes

$$a_k = \frac{c}{k^p}, \qquad p > \frac{1}{2},$$

and the random variable

$$X = \sum_{k=1}^{\infty} a_k \varepsilon_k.$$

We require $p > \frac{1}{2}$ because the sum of the squared step sizes must be finite:

$$\sum_{k=1}^{\infty} a_k^2$$

$$= c^2 \sum_{k=1}^{\infty} \frac{1}{k^{2p}} < \infty.$$

This follows from the p-series test, which states that $\sum 1/k^q$ converges when $q > 1$. Here, $q = 2$, so $p > \frac{1}{2}$.

Each term $a_k \varepsilon_k$ has mean zero because the two possible signs are equally likely. The terms are also independent, and each has variance a_k^2. Therefore, Kolmogorov's convergence theorem guarantees that the infinite series converges almost surely, meaning it converges with probability (1).

1. Spread and Variance

Because the terms are independent and have mean zero, the mean and variance of (X) are

$$\mathbb{E}[X] = 0,$$

$$\begin{aligned} &\operatorname{Var}(X) \\ &= \sum_{k=1}^{\infty} a_k^2 \\ &= c^2 \zeta(2p) \\ &= \sigma^2, \end{aligned}$$

where ζ is the Riemann zeta function and σ is the standard deviation.

We can choose any desired positive standard deviation by setting

$$\boxed{c = \frac{\sigma}{\sqrt{\zeta(2p)}}}.$$

This value is finite because (p>1/2) ensures that $\zeta(2p)$ is finite and positive. Changing c scales every step equally, controlling the overall spread without changing the independence of the coin flips.

For example, when $a_k = 1/k$, we have c=1 and p=1. Since $\zeta(2) = \pi^2/6$,

$$\mathrm{Var}(X) = \frac{\pi^2}{6}.$$

In a corresponding two-dimensional random walk with step lengths 1/k and independent, uniformly random directions, the expected squared distance from the origin is also $\pi^2/6$. Its root-mean-square distance is therefore

$$\sqrt{\mathbb{E}[|S_\infty|^2]} = \frac{\pi}{\sqrt{6}} \approx 1.28.$$

2. Bounds and Approximation Error

If we stop the walk after (N) steps, the position is

$$X_N = \sum_{k=1}^{N} a_k \varepsilon_k.$$

Since each sign has absolute value (1), the triangle inequality gives

$$\boxed{|X_N| \le \sum_{k=1}^{N} a_k}.$$

This is a guaranteed bound for every finite walk.

The difference between the finite walk and its limiting value is

$$X - X_N = \sum_{k=N+1}^{\infty} a_k \varepsilon_k.$$

Independence and the zero means eliminate the cross terms when calculating the squared error, giving

$$\begin{aligned}
&\mathbb{E}[(X - X_N)^2] \\
&\quad = \sum_{k=N+1}^{\infty} a_k^2 \\
&= c^2 \sum_{k=N+1}^{\infty} \frac{1}{k^{2p}}.
\end{aligned}$$

Because the full series of squared step sizes converges, this remaining sum approaches zero as (N) increases. Thus, the finite walk approximates the limiting random variable increasingly well in mean square.

There is an important distinction between finite and infinite walks. For $p > 1$, the sum of the step lengths $\sum 1/k^p$ is finite, so the infinite walk also has a deterministic bound. For $1/2 < p \leq 1$, the sum of the step lengths diverges, so the triangle inequality does not provide a finite bound for the infinite walk, even though it still converges almost surely.

We can also bound the probability of obtaining a large value. Hoeffding's inequality applies because the terms are independent and bounded. For the finite sum,

$$\begin{aligned}
&\Pr(|X_N| \geq t) \\
&\quad \leq \\
&2 \exp\left(-\frac{t^2}{2 \sum_{k=1}^{N} a_k^2} \right).
\end{aligned}$$

Since the finite sum of squared step sizes is at most σ^2, the limiting random variable satisfies

$$\boxed{\begin{aligned}&\Pr(|X| \geq t)\\&\leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right)\end{aligned}}.$$

This inequality shows that large deviations from zero become increasingly unlikely.

3. Shape of the Distribution

The distribution of (X) is symmetric about zero because every random sign is equally likely to be positive or negative. Depending on the step sizes, the distribution can appear uniform, bell-shaped, or somewhere between these shapes.

The central limit theorem helps explain when the distribution approaches a normal distribution. A key condition is that no individual term dominates the total variance. The fraction of the variance contributed by the first step is

$$\frac{a_1^2}{\sum_{k=1}^{\infty} a_k^2} = \frac{1}{\zeta(2p)}.$$

For p=1, this fraction is

$$\frac{6}{\pi^2} \approx 61\%$$

Thus, the first step contributes a large share of the total variance, helping explain why the distribution may differ substantially from a normal distribution.

We can measure this difference using excess kurtosis, which describes an aspect of a distribution's shape relative to a normal distribution. Each random sign has a fourth cumulant of -2, and cumulants add for independent random variables. Consequently,

$$\boxed{\kappa = -\frac{2\zeta(4p)}{\zeta(2p)^2}}.$$

A normal distribution has excess kurtosis 0, while a negative value indicates lighter tails and a flatter overall shape.

Step sizes (a_k)	Distribution's shape	Excess kurtosis
2^{-k}	Uniform on ([-1,1])	-1.2
k^{-1}	Relatively concentrated, bell-shaped	-0.8
$k^{-3/4}$	More rounded bell shape	About -0.35
k^{-p}, as $p \to \frac{1}{2}^{+}$	Approaches a normal shape after standardization	$\to 0$

As) approaches 1/2 from above, $\zeta(2p)$ grows without bound, while (\zeta(4p)) approaches the finite value (\zeta(2)). Therefore,

$$\kappa = -\frac{2\zeta(4p)}{\zeta(2p)^2} \longrightarrow 0.$$

At the same time, the fraction of variance contributed by each fixed early step approaches zero. This means the total variance is spread across increasingly many small contributions. More formally, the normalized sum X/σ converges in distribution to a standard normal random variable as $p \to 1/2^{+}$, because no single step contributes a substantial fraction of the total variance.

The boundary p=1/2 is excluded because the sum of squared step sizes diverges there, so the convergence theorem used above no longer guarantees convergence.

A special case occurs when $a_k = 2^{-k}$. To see why the distribution is uniform, define

$$B_k = \frac{\varepsilon_k + 1}{2}.$$

Each B_k is an independent binary digit, equally likely to be 0 or 1. We can rewrite the sum as

$$X = 2\sum_{k=1}^{\infty} \frac{B_k}{2^k} - 1.$$

The binary expansion $\sum B_k/2^k$ is uniformly distributed on [0,1]. Therefore, the transformation $U \mapsto 2U - 1$ produces a uniform random variable on [-1,1].

4. Generating Random Numbers

The walk can generate random values with a chosen standard deviation and distribution shape.

1. Choose a desired standard deviation $\sigma > 0$ and an exponent p>1/2.
2. Set $c = \sigma/\sqrt{\zeta(2p)}$.
3. Generate N independent fair coin flips, assigning each flip a value $\varepsilon_k = +1$ or -1.
4. Calculate

$$X_N = \sum_{k=1}^{N} c, k^{-p}\varepsilon_k.$$

The output has mean zero and variance

$$\begin{aligned} &\mathrm{Var}(X_N) \\ &= c^2 \sum_{k=1}^{N} \frac{1}{k^{2p}}, \end{aligned}$$

which approaches σ^2 as N increases. Its value is also guaranteed to satisfy

$$|X_N| \le c\sum_{k=1}^{N} \frac{1}{k^p}.$$

The variance approaches the desired value, although a finite number of steps does not generally produce exactly the full target variance.

Instead of using positive and negative signs, choose independent angles θ_k uniformly from $[0, 2\pi)$. Each step is

$$\begin{aligned}&\mathbf{S}_N\\&= \sum_{k=1}^{N} c, k^{-p}\\&(\cos\theta_k, \sin\theta_k).\end{aligned}$$

Uniform angles ensure that each step has mean vector zero. Since each step has squared length a_k^2 and the steps are independent,

$$\begin{aligned}&\mathbb{E}[|S_\infty|^2]\\&= \sum_{k=1}^{\infty} a_k^2\\&= c^2\zeta(2p).\end{aligned}$$

The distribution is rotationally symmetric because rotating all the angles by the same amount preserves their probability distribution. Each coordinate has variance

$$\begin{aligned}&\operatorname{Var}(S_{\infty,x})\\&= \operatorname{Var}(S_{\infty,y})\\&= \frac{1}{2}c^2\zeta(2p).\end{aligned}$$

The distance from the origin, $|S_\infty|$, has a nonnegative, generally right-skewed distribution, as illustrated by the paper's second histogram.

The random walk can also generate values from a different target distribution by transforming its output. Let F be the continuous cumulative distribution function (CDF) of X. The probability integral transform states that

$$U = F(X)$$

is uniformly distributed on ([0,1]).

If G is the CDF of a desired distribution, define

$$\boxed{Y = G^{-1}(F(X))},$$

where G^{-1} is the target distribution's quantile function. Then Y follows the distribution G. This method allows the same random walk to generate values from many distributions through an appropriate transformation.

A decreasing-step random walk can act as a random number generator because its independent, mean-zero steps have a finite total variance whenever p>1/2. This condition guarantees almost-sure convergence and allows us to calculate the limiting variance. The step sizes determine the distribution's spread and shape, while probability inequalities describe the likelihood of large values. By selecting p, scaling the steps with c, extending the walk to two dimensions, or transforming the output, we can use this process to generate a wide variety of random values.

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